Convertible Bond Pricing with Longstaff-Schwartz

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Abstract: Convertible bonds are hybrid securities that combine features of debt and equity, making their valuation challenging due to the embedded optional conversion. This paper applies the Longstaff–Schwartz least-squares Monte Carlo (LSMC) algorithm to price a convertible bond, to provide an exposition of the method's implementation and behaviour. Monte Carlo simulation with a regression-based optimal stopping strategy offers flexibility in handling the American-style conversion option and potential path-dependent features, in contrast to lattice or finite-difference approaches which struggle with complex call or put schedules and multiple state variables.

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1 Introduction

Convertible bonds are hybrid securities that offer an option to convert the bond into a predetermined number of shares of the issuer's stock. In other words, a convertible bond is a corporate bond with a call option on the company's stock. If the company's stock performs well, the bondholder can convert into shares and benefit from the equity appreciation; if the stock performs poorly, the investor can hold the bond and receive regular interest and the face value at maturity.

Every convertible bond has certain key parameters that determine its value and conversion mechanics. The *face value* is the principal amount of the bond that will be returned to the investor at maturity if not converted. Additionally, the *conversion ratio* is the number of shares of stock that the bondholder is entitled to receive upon converting one bond. From these definitions, we define the *conversion price* as the effective price per share at which conversion happens. If we let the bond's face value be denoted as F and N is the conversion ratio, then conversion price is defined as

$$\mathcal{P} = \frac{F}{N}.$$

Another important metric is the *conversion value*, also known as *parity*, which is the current market value of the shares into which the bond can be converted. Mathematically speaking, parity $= S \times N$, where S is the current stock price. It represents the value the bondholder would get by converting the bond at the prevailing stock price. On the other hand, the bond's *straight value* is the value of the bond without the conversion option. Essentially, the present value of its promised cash flows (coupons plus principal) discounted at the yield of a comparable non-convertible bond. The straight value (also known as *bond floor*) represents a lower bound for the convertible's price, since it reflects what the bond would be worth as a pure debt instrument. Hence, a convertible's market price is composed of two components: the straight bond value and the option component. The *conversion premium* is the amount by which the convertible's market price exceeds its conversion value (parity). Because the investor can always choose not to convert and instead hold the bond to maturity, the convertible's price should never fall below the greater of its straight-bond value and its conversion value. Arbitrage arguments guarantee that a convertible bond will trade at at least the higher of its bond floor and its parity value. If it were to trade below that level, an arbitrage opportunity exists, and one could buy the underpriced convertible and either hold it as a bond or convert it to stock to realise an instant profit.[1] Hence, the minimum value of a convertible at any time is max(straight bond value, conversion value).

Since convertibles consist of a stock option, their value and behaviour depend on the *moneyness* of the conversion option. The conversion option is considered *in-the-money* (ITM) if the current stock price exceeds the conversion price, i.e., $S > \mathcal{P} \iff SN > F$. Conversely, it is *out-of-the-money* (OTM) if the stock price is below the conversion price, $S < \mathcal{P}$, making immediate conversion irrational. If $S \approx \mathcal{P}$, then the option is near *at-the-money* (ATM). When the option is deeply ITM ($S \gg \mathcal{P}$), the convertible bond's price is largely driven by its equity value, and it behaves like an equity-like instrument. Here, the bond's market price will be close to the conversion value and will move nearly one-for-one with the underlying stock, with only a small premium reflecting any remaining

[1]Mathematical proof in Appendix 1

bond interest or optionality. In the opposite case, when the option is far OTM ($S \ll \mathcal{P}$), the convertible behaves like a bond-like instrument. Here, the stock price is so low that conversion is unlikely, so the convertible's value moves toward its bond floor and is more sensitive to interest rates and credit risk than to the stock price. Finally, the case at ATM is often referred as a *balanced convertible*, where the stock price is in around the conversion price ($S \approx \mathcal{P}$) such that the convertible has significant bond value and significant option value at the same time. in this region, the convertible's price will consist of bond characteristics as well as equity option upside, meaning it reacts to both the underlying movements and interest rate/credit changes.

In extreme situations where the issuer's stock price has fallen very sharply and the company's credit has dropped, a convertible can trade at *distressed* levels. Distressed convertible is a bond whose conversion option is practically worthless (the stock is deeply OTM), such that the issue trades almost purely on its remaining bond value. These convertibles often exhibit high yields, due to the issuer's elevated credit risk, and a significantly high conversion premium, since the market price of the bond significantly exceeds the negligible equity conversion value. Hence, a distressed convertible is one that has become debt-like in value due to a collapsed stock price and investor concerns about the issuer's solvency. Under these circumstances, the bond may trade at a substantial discount, but it also means that if the company improves and the stock rebounds, the convertible has a lot of upside potential from those low levels.

Due to the nature of convertibles, it is rare to find European-style convertible bonds. Realistically, convertible bonds are structured with American-style (or Bermudan-style) derivative, whereby the user can convert at any point between the issue of the bond and the maturity date. American-style derivatives present more challenge than a European-style derivative to price, since one cannot take the fair value of the derivative at maturity. Instead, at each point during the lifetime of the derivative, one must compare the value of immediate exercise with the expected value of holding the derivative, with the opportunity to exercise later. Traditional valuation approaches for convertibles proceed either by discrete-time lattices. In a binomial or trinomial lattice the stock process is discretised, and at each node one solves a backward induction with an early-exercise comparison. This method as well as other such as finite-difference methods encounter practical issues for realistic contracts. For example, callable schedules, soft-call provisions, coupon accrual conventions, and defaultable cashflow legs induce path dependence. Stochastic interest rates and stochastic credit spreads introduce additional state variables, and equity dynamics may include jumps or local stochastic volatility. Consequently, classical grid methods can become impractical in American-style optionality coupled with multiple stochastic drivers and event-driven payoffs.

Monte Carlo simulation can help with dimensionality and accommodates path dependence, however a naïve simulation of discount payoffs does not resolve the optimal stopping inherent in conversion. Formally, if we let $(\mathcal{F}_t)_{t\in[0,T]}$ denote the market filtration and $\mathbb{Q}$ the risk-neutral measure, then the convertible's value at time t solves the Snell envelope representation

$$V_t = \operatorname*{ess\,sup}_{\tau\in\mathcal{T}_{t,T}} \mathbb{E}_{\mathbb{Q}}\left[e^{-\int_t^\tau r_u du} X_\tau \middle|\, \mathcal{F}_t \right],$$

where $\mathcal{T}_{t,T}$ is the set of $\mathcal{F}_t$ stopping times taking values in $[t, T]$, r is the short rate, and X_τ is the cashflow at the first event (conversion, call, default, or maturity) occurring at τ. A forward simulation samples $\{X_\tau\}$ given a fixed policy τ, and as such, it does not learn the maximising policy. What is missing is a way to estimate, from cross-sectional simulation data, the conditional expectation of continuation that defines the optimal policy through the dynamic programming principle.

The *Longstaff-Schwartz least-squares Monte Carlo* (LSM) method addresses this by using least-squares to estimate the conditional expected value of continuing to hold the derivative at each time step. On a discrete grid $0 = t_0 < t_1 < ... < t_n = T$, where n denotes number of time steps, define the continuation value at time t_i by

$$C(t_i, S_{t_i}) = \mathbb{E}_{\mathbb{Q}}\left[e^{-\int_{t_i}^{t_{i+1}} r_u du} V_{t_{i+1}} \mid S_{t_i}\right],$$

where S is the equity state.

In this paper, we use the algorithm to price different types of convertibles. We begin by formalising conditional expectation as an L^2 projection and deriving least-squares estimators from the projection, before presenting the LSM algorithm and applying it to a non-callable, non-puttable, zero-coupon convertible under geometric Brownian motion, and a callable, defaultable convertible in which issuer call is modelled as a forced stopping at call dates and default is incorporated either by intensity-based arrival or by stock-triggered default thresholds with recovery, demonstrating how calls and credit risk reshape the continuation region and compress upside. Finally we document the LSM results and compare them to finite-difference methods.

2 Conditional Expectations and Least-Squares Regression

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. For a sub-sigma-algebra $\mathcal{G} \subseteq \mathcal{F}$, the conditional expectation of an integrable random variable X with respect to $\mathcal{G}$, denoted $\mathbb{E}[X \mid \mathcal{G}]$, is the best estimate of X given the information $\mathcal{G}$. In fact $\mathbb{E}[X \mid \mathcal{G}]$ is the unique $\mathcal{G}$-measurable random variable that minimizes the mean squared error: $\mathbb{E}[(X-Y)^2]$ among all $\mathcal{G}$-measurable Y. In other words,

$$\mathbb{E}[(X - \mathbb{E}[X|\mathcal{G}])^2] \leq \mathbb{E}[(X-Y)^2], \quad Y \text{ that is } \mathcal{G}\text{-measurable.}$$

Equivalently, one has the orthogonality condition: $\mathbb{E}[(X - \mathbb{E}[X|\mathcal{G}])Z] = 0$ for all $\mathcal{G}$-measurable Z.

The link between conditional expectation and least-squares regression is fundamental to the LSM method. Consider two square-integrable random variables X and Y. The conditional expectation $\mathbb{E}[Y \mid X]$ is a function of X (measurable w.r.t. X) which can be thought of as the best predictor of Y given X. In fact, we can prove that $\mathbb{E}[Y \mid X]$ is the unique function of X that achieves the minimum mean squared error $\mathbb{E}[(Y - f(X))^2]$ among all measurable functions f. In other words,

$$\mathbb{E}[(Y - \mathbb{E}[Y \mid X])^2] \leq \mathbb{E}[(Y - f(X))^2], \quad f \text{ measurable with } \mathbb{E}[f(X)^2] < \infty. \tag{1}$$

We can prove this by looking at the right hand side of (1):

$$\begin{aligned}\mathbb{E}[(Y - f(X))^2] &= \mathbb{E}[(Y - f(X) + \mathbb{E}[Y \mid X] - \mathbb{E}[Y \mid X])^2] \\ &= \mathbb{E}[((Y - \mathbb{E}[Y \mid X]) - (f(X) - \mathbb{E}[Y \mid X]))^2] \\ &= \mathbb{E}[(Y - \mathbb{E}[Y \mid X])^2 + (f(X) - \mathbb{E}[Y \mid X])^2 - 2(Y - \mathbb{E}[Y \mid X])(f(X) - \mathbb{E}[Y \mid X])],\end{aligned}$$

and by linearity of expectations,

$$= \mathbb{E}[(Y - \mathbb{E}[Y \mid X])^2] + \mathbb{E}[(f(X) - \mathbb{E}[Y \mid X])^2] - 2\,\mathbb{E}[(Y - \mathbb{E}[Y \mid X])(f(X) - \mathbb{E}[Y \mid X])].$$

If we look at the final term, we can apply the tower property to get,

$$\begin{aligned}\mathbb{E}[(Y - \mathbb{E}[Y \mid X])(f(X) - \mathbb{E}[Y \mid X])] &= \mathbb{E}[\mathbb{E}[(Y - \mathbb{E}[Y \mid X])(f(X) - \mathbb{E}[Y \mid X])] \mid X] \\ &= \mathbb{E}[\mathbb{E}[Yf(X) - Y\,\mathbb{E}[Y \mid X] - f(X)\,\mathbb{E}[Y \mid X] + (\mathbb{E}[Y \mid X])^2] \mid X]\end{aligned}$$

Using the property that $\mathbb{E}[g(X)Y \mid X] = g(X)\,\mathbb{E}[Y \mid X]$:

$$\begin{aligned}&= \mathbb{E}[f(X)\,\mathbb{E}[Y \mid X] - (\mathbb{E}[Y \mid X])^2 - f(X)\,\mathbb{E}[Y \mid X] + (\mathbb{E}[Y \mid X])^2] \\ &= \mathbb{E}[f(X)\,\mathbb{E}[Y \mid X] - f(X)\,\mathbb{E}[Y \mid X]] = 0.\end{aligned}$$

Hence,

$$\mathbb{E}[(Y - f(X))^2] = \mathbb{E}[(Y - \mathbb{E}[Y \mid X])^2] + \mathbb{E}[(f(X) - \mathbb{E}[Y \mid X])^2] - 2 \times 0.$$

Since, $\mathbb{E}[(f(X) - \mathbb{E}[Y \mid X])^2] \geq 0$,

$$\mathbb{E}[(Y - f(X))^2] \geq \mathbb{E}[(Y - \mathbb{E}[Y \mid X])^2].$$

In other words,

$$\mathbb{E}[Y \mid X] = \arg\min_{f(X)} \mathbb{E}[(Y - f(X))^2]. \tag{2}$$

In practice, one approximates this optimal function by restricting to a class of functions (such as polynomials in X or other basis functions) and performing a least-squares regression on historical data. Suppose we choose finite set of basis functions $\{\phi_1(x), \phi_2(x), ..., \phi_m(x)\}$, for example powers of x, $\{1, x, x^2, ..., x^m\}$ to span an approximation space. We then approximate conditional expectation of the form

$$f(X) = \alpha_1\phi_1(X) + \alpha_2\phi_2(X) + ... + \alpha_m\phi_m(X) = \sum_{j=1}^{m} \alpha_j\phi_j(X),$$

that best fits Y in the least-squares sense (Longstaff et al., 2001). The regression problem is to find coefficients $\boldsymbol{\alpha} = (\alpha_1, ..., \alpha_m)$ minimising

$$R(\boldsymbol{\alpha}) = \mathbb{E}\left[\left(Y - \sum_{j=1}^{m} \alpha_j\phi_j(X)\right)^2\right].$$

We can write the residual as $\epsilon(\boldsymbol{\alpha}) = Y - \boldsymbol{\alpha}^\top\boldsymbol{\phi}(X)$, then

$$\begin{aligned}R(\boldsymbol{\alpha}) &= \mathbb{E}[\epsilon(\boldsymbol{\alpha})^2] = \mathbb{E}[(Y - \boldsymbol{\alpha}^\top\boldsymbol{\phi}(X))^2] \\ &= \mathbb{E}[Y^2 - 2Y\boldsymbol{\alpha}^\top\boldsymbol{\phi}(X) + (\boldsymbol{\alpha}^\top\boldsymbol{\phi}(X))^2],\end{aligned}$$

by linearity

$$R(\boldsymbol{\alpha}) = \mathbb{E}[Y^2] - 2\,\mathbb{E}[Y(\boldsymbol{\alpha}^\top\boldsymbol{\phi}(X))] + \mathbb{E}[(\boldsymbol{\alpha}^\top\boldsymbol{\phi}(X))^2].$$

By using the fact that $(A^\top B)(B^\top A) = A^\top(BB^\top)A$ and linearity,

$$R(\boldsymbol{\alpha}) = \mathbb{E}[Y^2] - 2\boldsymbol{\alpha}^\top\,\mathbb{E}[\boldsymbol{\phi}(X)Y] + \boldsymbol{\alpha}^\top\,\mathbb{E}[\boldsymbol{\phi}(X)\boldsymbol{\phi}(X)^\top]\boldsymbol{\alpha}.$$

Define $\varphi := \mathbb{E}[\boldsymbol{\phi}\boldsymbol{\phi}^\top]$, $b := \mathbb{E}[\boldsymbol{\phi}(X)Y]$, and $c := \mathbb{E}[Y^2]$, then the risk is the quadratic form $R(\boldsymbol{\alpha}) = c - 2\boldsymbol{\alpha}^\top b + \boldsymbol{\alpha}^\top \varphi \boldsymbol{\alpha}$. We can note that c, φ, b do not depend on $\boldsymbol{\alpha}$ and b, φ depend on the joint law of (X, Y) and the choice of basis. Differentiating $R(\boldsymbol{\alpha})$ with respect to $\boldsymbol{\alpha}$ gives us

$$\nabla_{\boldsymbol{\alpha}} R(\boldsymbol{\alpha}) = -2b + 2\varphi\boldsymbol{\alpha}.$$

Setting this to zero gives us $\varphi\boldsymbol{\alpha} = b$. This linear system is the population normal equations. If the chosen basis functions are linearly independent in L^2 (equivalently, there is β such that $\beta^\top \phi(X) = 0$ a.s.), then φ is invertible and positive definite, hence

$$\boldsymbol{\alpha} = \varphi^{-1} b.$$

In the context of the Longstaff-Schwartz method, Y will represent the realised future payoff from continuation of an option, and X will be the state variable(s) at the current time (such as the underlying stock price). By regressing Y on functions of X across many simulated paths, we can obtain an estimate of the continuation value, $\mathbb{E}[Y \mid X]$, without needing an analytic formula.

3 Least-Squares Monte Carlo (LSM) for Convertible Bond Pricing

Convertible bonds in practice can be converted into the predetermined shares at any point of its lifetime, hence acting as an American-style derivative. Due to the complexity of modelling continuous exercise time, we will model convertible bonds under a *Bermudan-style derivative*, which allows us to exercise the conversion over a fixed and finite number of times up to and including maturity. We will split this section into two parts: the first is an introduction to LSM for a Bermudan option. The second part will be about LSM for convertible bonds.

3.1 Bermudan Option

Say we have a Bermudan option V with $n \in \mathbb{N}$ exercise times $0 < t_1 < t_2 < ... < t_n = t$, then define the immediate exercise value for a Bermudan at t_i as

$$h_i(S_{t_i}) := (S_{t_i} - K_i)^+,$$

where K is the strike and S is the underlying. The underlying price, V_{underl}, of the Bermudan option at time t_i is

$$V_{\text{underl}}(t_i) = \max\left(h_i(S_{t_i}),\ e^{-r(t_{i+1}-t_i)}\, \mathbb{E}_{\mathbb{Q}}[V_{t_{i+1}} \mid \mathcal{F}_{t_i}]\right).$$

As the exercise mesh becomes desne (i.e., $\max_i(t_{i+1} - t_i) \to 0$), the Bermudan-style derivative converges to an American derivative. For the paths where the derivative is ITM, we only exercise if and only if the exercise is more profitable than continuation. Hence, to calculate the fair price at time t_i ($i \neq n$) of a Bermudan-style derivative, we take the maximum value of either exercising at time t_i or hold to exercise at t_{i+1}. In other words, given the filtration $\{\mathcal{F}_{\sqcup_\rangle}\}_{i\in[0,n]}$,

$$V(t_i, ..., t_n; t_i) = \max(V(t_{i+1}, ..., t_n; t_i), V_{\text{underl}}(t_i)), \tag{3}$$

where,

$$V(t_{i+1},...,t_n;t_i) = e^{-r(t_{i+1}-t_i)}\,\mathbb{E}\left[V_{t_{i+1}}|\mathcal{F}_{t_i}\right]. \tag{4}$$

Paths that are ATM or OTM are automatically not profitable to exercise at time t_i and the investor will choose to hold the derivative instead. Hence the value of ATM and OTM the derivative at time t_i will be the same value as the derivative at time t_{i+1} discounted back to t_i.

To calculate the conditional expectation in $V(t_{i+1},...,t_n;t_i)$ is not obvious, since we have multiple timestamps that we can exercise; hence not a simple case of using a Black-Scholes for example. As such, we do not have a real representation of the discounted future cashflow.

One method that was designed to tackle this was through resimulation, using Monte Carlo. The straightforward way to calculate the conditional expectation is to create in t_i a new MC simulation (conditional) on each path. This leads to a much higher number of total simulated paths needed. If one considers $n > 1$, then this method becomes impractical. The required number of paths grows exponentially by $(n_{\text{paths}})^{n_{\text{exercise}}}$.

A solution to this could be to use *binning*. For example, say we had a simple Bermudan option which has two exercise dates, and in t_1 we want to evaluate the option of receiving $e^{-rt_1}(S_{t_1}-K_1)$ in t_1 or receiving $e^{-r(t_2-t_1)}(S_{t_2}-K_2)^+$ at time t_2, where K_1, K_2 are given strikes. The optimal exercise in t_1 compares the exercise value with the value of the t_2 option:

$$e^{-r(t_2-t_1)}\,\mathbb{E}\left[V_{t_2}|\mathcal{F}_{t_1}\right].$$

If we take the risk-neutral stock price to be modelled under a geometric Brownian motion:

$$dS_t = rS_t dt + \sigma S_t dW_t,$$

it is then sufficient to calculate

$$e^{-r(t_2-t_1)}\,\mathbb{E}_{\mathbb{Q}}\left[(S_{t_2}-K_2)^+ \mid S_{t_1}\right].$$

We can then use binning to calculate an approximation to the conditional expectation. For a given S_{t_1} none or very few values of the continuation values exist. An estimate is not possible or else exhibits a *foresight* bias, i.e, non-measurable bias. For an interval $[S_1-\epsilon, S_1+\epsilon]$ with sufficiently large ϵ we have enough values to estimate

$$e^{-r(t_2-t_1)}\,\mathbb{E}_{\mathbb{Q}}\left[(S_{t_2}-K_2)^+ \mid S_{t_1}\in[S_1-\epsilon, S_1+\epsilon]\right],$$

which in turn may be used as an estimate for

$$e^{-r(t_2-t_1)}\,\mathbb{E}_{\mathbb{Q}}\left[(S_{t_2}-K_2)^+ \mid S_{t_1}=S_1\right].$$

Now consider the binning as an estimate of the conditional expectation $e^{-r(t_2-t_1)}\,\mathbb{E}_{\mathbb{Q}}\left[V_{t_2}\mid Z\right]$. We can calculate the conditional expectation:

$$H := e^{-r(t_2-t_1)}\,\mathbb{E}_{\mathbb{Q}}\left[V_{t_2}\mid Z\in U_i\right] = e^{-r(t_2-t_1)}\,\mathbb{E}_{\mathbb{Q}}\left[V_{t_2}\mid Z^{-1}(U_i)\right]$$

where U_i are the disjoint discretisation bins of the set of values of the state space $Z(\Omega)$. By taking the lemma (1) defined in section 2, we write this as a minimisation problem for the vector $(H_i)_{i\in[1,n]}$ for disjoint bins U_i:

$$\sum_i \mathbb{E}_{\mathbb{Q}}\left[\left(e^{-r(t_2-t_1)}V_{t_2}-H_i\right)^2 \mid Z\in U_i\right] = \min_{G\in\mathbb{R}}\sum_i \mathbb{E}_{\mathbb{Q}}\left[\left(e^{-r(t_2-t_1)}V_{t_2}-G\right)^2 \mid Z\in U_i\right]. \tag{5}$$

In fact, if we let $\mathcal{H}$ be the function space $H : \mathbb{R} \to \mathbb{R}$ being constant on the bins $Z^{-1}(U_i)$, and let $H \in \mathcal{H}$ with $H(\omega) := H_i$ for $\omega \in Z^{-1}(U_i)$, then (5) is equivalent to

$$\mathbb{E}_{\mathbb{Q}}\left[\left(e^{-r(t_2-t_1)}V_{t_2} - H\right)^2 \mid Z\right] = \min_{G\in\mathbb{R}} \mathbb{E}_{\mathbb{Q}}\left[\left(e^{-r(t_2-t_1)}V_{t_2} - G\right)^2 \mid Z\right].$$

The problem about binning stems from the partitioning of the state space $Z(\Omega)$ into finite number of bins. This results in a piecewise constant approximation of the conditional expectation. We can solve this issue by approximating the conditional expectation using a smooth function of the state variable Z. We can do this via a *least-squares approximation* of the conditional expectation.

Let $(\Omega, \mathcal{F}, \mathbb{Q}, \{\mathcal{F}_t\})$ be a filtrated probability space and V on $\mathcal{F}_{t_i}$-measurable random variable defined as the conditional expectation of U:

$$V = \mathbb{E}_{\mathbb{Q}}[U \mid \mathcal{F}_{t_1}],$$

where U is at least $\mathcal{F}$-measurable. Furthermore let $Y := (Y_1, ..., Y_p)$ be a given $\mathcal{F}_{t_i}$-measurable random variable and $f : \mathbb{R}^p \times \mathbb{R}^q$ be a given function. Let $\Omega^* = \{\omega_1, ..., \omega_m\}$ be drawing from Ω and $\alpha^* := (\alpha_1, ..., \alpha_q)$ such that:

$$|| \; U - f(Y, \alpha^*) \; ||_{L_2(\Omega)} = \min_{\alpha} || \; U - f(Y, \alpha) \; ||_{L_2(\Omega)},$$

where

$$|| \; U - f(Y, \alpha^*) \; ||^2_{L_2(\Omega^*)} = \sum_{j=1}^{m} (U(\omega_j) - f(Y(\omega_j), \alpha^*))^2.$$

If we set $V^{LS} := f(Y, \alpha^*)$, then V^{LS} is $\mathcal{F}_{\sqcup_\rangle}$-measurable and is defined over Ω. Additionally, it is a least-square approximation of V on Ω^*. Mathematically speaking, if we let $\Omega^* = \{\omega_1, ..., \omega_m\}$ be a given sample space with the given random variables $V : \Omega^* \to \mathbb{R}$ and $Y := (Y_1, ..., Y_p) : \Omega^* \to \mathbb{R}^p$. Furthermore, let

$$f(y_1, ..., y_p, \alpha_1, ..., \alpha_p) := \sum_{i}^{p} \alpha_i y_i,$$

be the Longstaff-Schwartz function to represent our basis function of the conditional expectation. Then we have for any α^* with $X^\top X \alpha^* = X^\top v$,

$$|| \; V - f(Y, \alpha^*) \; ||_{L_2(\Omega^*)} = \min_{\alpha} || \; V - f(Y, \alpha) \; ||_{L_2(\Omega^*)},$$

where,

$$X := \begin{pmatrix} Y_1(\omega_1) & \cdots & Y_p(\omega_1) \\ \vdots & \ddots & \vdots \\ Y_1(\omega_m) & \cdots & Y_p(\omega_m) \end{pmatrix} \qquad v := \begin{pmatrix} V(\omega_1) \\ \vdots \\ V(\omega_m) \end{pmatrix}.$$

If $(X^\top X)^{-1}$ exists then we get a final solution $\alpha^* := (X^\top X)^{-1} X^\top v$, which is the conditional expectation for holding a Bermudan option to t_{i+1}.

3.2 Convertible Bond

Unlike Bermudan options, convertible bonds have interim cashflows (coupons), which means that we must change the discounted value from $V_{T_{i+1}}$ to $V_{T_{i+1}} + c_{i+1}$, for some coupon c_{i+1}. When considering the exercise we take the conversion at time t_i to be $X_i^{\text{conv}} = NS_{t_i}$, giving us

$$V_{t_i} = \max(NS_{t_i}, C(t_i, S_{t_i})), \qquad C(t_i, S_{t_i}) = \mathbb{E}_{\mathbb{Q}}[Y_{i+1} \mid S_{t_i}], \qquad Y_{i+1} = e^{-r\Delta t_i}(V_{t_{i+1}} + c_{i+1}).$$

We can noted that the assumption for the continuation value $C(t_i, S_{t_i})$ is the underlying follows a Markov process and is measurbale at t_i. We estimate the conditional expectation $C(t_i, S_{t_i}) = \mathbb{E}_{\mathbb{Q}}[Y_{i+1} \mid S_{t_i}]$ by regressing the realised discounted cashflows $Y_{i+1}^{(k)}$ on a *Laguerre series* to the pair $(S_{t_i}, C_{t_{i+1}})$. This pair corresponds to (X, Y) respectively in the least-squares minimisation explanation in section 2, and (V, X) in our Bermudan option example. We minimise the least-squares for the future cashflow at t_{i+1} to get the best estimate of the continuation value $C(t_i, S_{t_i})$. Like with the Bermudan option, we want to represent $C(t_i, S_{t_i})$ as a basis function. This basis function is square-integrable as we are minimising the mean squared error, which only well-defined and finite if the second moment exist. In other words, we want this conditional expectation to be an approximate L^2 projection, hence want the basis function to be of L^2. In practice, a common option to use is the *Laguerre Polynomials*, given as

$$L_n(x) = \sum_{k=0}^{n} (-1)^k \binom{n}{k} \frac{x^k}{k!}.$$

We can represent the conditional value as

$$C(t_i, S_{t_i}) \approx \sum_{j=0}^{\infty} \alpha_j L_j \left(\frac{S_{t_i}}{\mathcal{P}} \right),$$

for α_j and S_{t_n} to be determined. For some degree $m \in \mathbb{N}$ of the Laguerre polynomials that we want to approximate the conditional expectation for, we can choose $\boldsymbol{\alpha} = (\alpha_0, ..., \alpha_m)$ that minimises the mean squared error

$$\boldsymbol{\alpha}^* = \underset{\boldsymbol{\alpha} \in \mathbb{R}^{m+1}}{\arg\min}\, \mathbb{E}_{\mathbb{Q}} \left[\left(Y_{i+1} - \sum_{j=0}^{m} \alpha_j L_j \left(\frac{S_{t_i}}{\mathcal{P}} \right) \right)^2 \right].$$

Under Monte Carlo simulation with simulated paths $k = 1, ..., M$, this population is represented by the sample least-squares problem

$$\hat{\boldsymbol{\alpha}}^* = \underset{\boldsymbol{\alpha} \in \mathbb{R}^{m+1}}{\arg\min} \sum_{k=1}^{M} \left(Y_{i+1}^{(k)} - \sum_{j=0}^{m} \alpha_j L_j \left(\frac{S_{t_i}^{(k)}}{\mathcal{P}} \right) \right)^2, \tag{6}$$

so that $\hat{C}(t_i, S_{t_i})$ becomes the fitted value from regressing the discounted future cashflows $Y_{i+1}^{(k)}$ on the Laguerre basis evaluated at $S_{t_i}^{(k)}$.

4 Results

4.1 Non-callable, Non-puttable Convertible Bond

We implement the ATM, and OTM using the rules set in equation (4) at time t, and to exercise if the bond is ITM and above the conversion price. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be our probability space with the filtration $\{\mathcal{F}_t\}_{t\in[0,T]}$, where T is maturity of the convertible bond. We will model our underlying process $\{S_t\}_{t\in[0,T]}$ as a Geometric Brownian Motion with respect to the probability space, with the risk-free asset, volatility, and dividend yields as r, σ, and q respectively. Under Euler's discretisation and risk-neutral settings, we can represent the underlying as

$$S_{t+\Delta t} = S_t + (r-q)S_t\Delta t + \sigma S_t\sqrt{\Delta t}Z, \qquad Z \sim \mathcal{N}(0,1).$$

Using the same variables defined in subsection 3.2, we can note that we exercise if $NS_{t_i} > C(t_i, S_{t_i}) \approx \hat{C}(t_i, S_{t_i})$, otherwise we hold to t_{i+1}, for $0 = t_0 < t_1 < ... < t_{n-1} < t_n = T$.

We calculate the value of the convertible bond by using backwards induction. We begin at time $t_n = T$. After the last exercise time, the value of the future payments is $C(t_n, S_{t_n}) = 0$. Our induction step we set $t = t_i$, which gives us:

$$\begin{cases} V_{t_i} = NS_{t_i}^{(k)}, & \text{if } NS_{t_i}^{(k)} \geq \hat{C}(t_i, S_{t_i}^{(k)}), \\ Y_{i+1}, & \text{otherwise.} \end{cases}$$

where $Y_{i+1} = e^{-r\Delta t_i}(V_{t_{i+1}}^{(k)} + c_{i+1})$, N is the number of predetermined shares upon exercise, and $c_{i+1} = 0$.[2]. It is important to note that $\hat{C}(t_i, S_{t_i}^{(k)}) \approx \text{Regress}\left(Y_{i+1}^{(k)} \text{ on basis}(S_{t_i}^{(k)})\right)$. Additionally, we calculate the value of $C(t_i, S_{t_i})$ through the regression problem given in (6),with the scaled regressor $S_{t_i}^{(k)}/\mathcal{P}$.

Finally, at time t_0 the convertible price is estimated by the Monte Carlo average

$$V_0 \approx \frac{1}{M}\sum_{k=1}^{M} V_{t_0}^{(k)}.$$

In the experiment, we consider a zero-coupon, non-callable, non-puttable convertible bond with face value $F = 100$ and conversion ratio $N = 1$. Hence the conversion price is $\mathcal{P} = F/N = 100$. The stock is initial set to $S_0 = 100$ such that it is at-the-money at interception, with parameters $(r, q, \sigma) = (0.04, 0, 0.1)$ and maturity $T = 2$ years. The LSM continuation is approximated using a Laguerre basis scaled by $S/\mathcal{P}$. Since for this here $q = 0$, early conversion is not economically incentivised by foregone cashflows since converting early replaces the convex claim on S, that being the embedded option-like component, with the linear payoff NS_t and hence sacrifices time value. In continuous-time setting, this referred to as the *no early exercise* idea for a non-dividend-paying American call. As a result, we expect most paths to delay conversion until maturity, and the price to be close to the European decomposition

$$V_0 = \mathbb{E}_{\mathbb{Q}}\left[e^{-rT}\max(NS_T, F)\right] = Fe^{-rT} + \mathbb{E}_{\mathbb{Q}}\left[e^{-rT}(NS_T - F)^+\right].$$

[2] zero coupon bond $c_{i+1} = 0, \forall i$

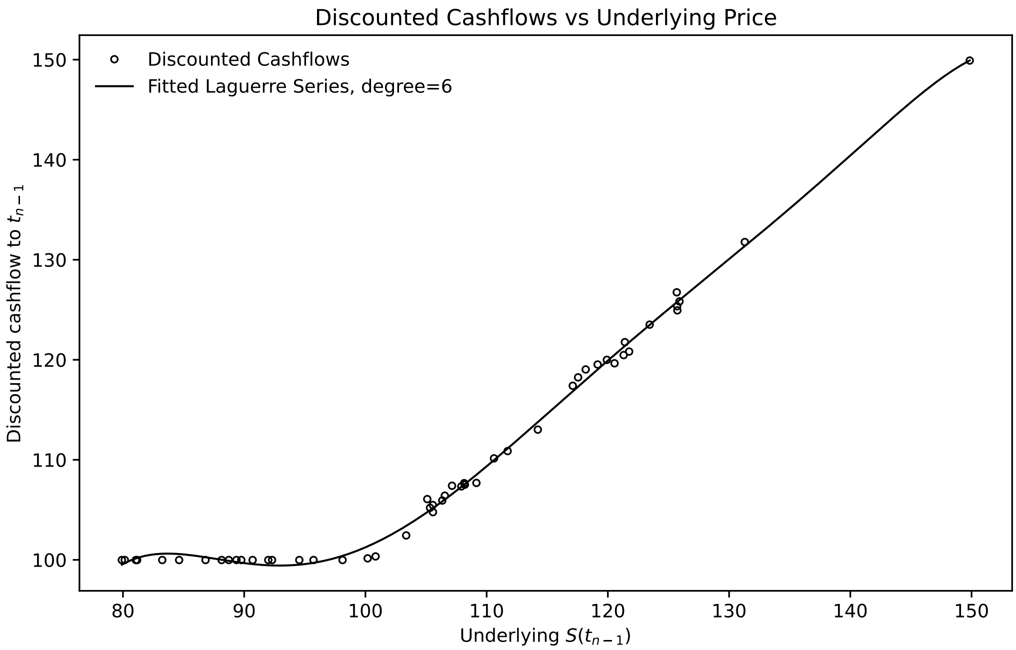

Figure 1: Discounted cashflows vs underlying price and Laguerre fit. Discounted realised cashflows at the penultimate time stop t_{n-1} plotted against $S_{t_{n-1}}$, with the fitted Laguerre-series approximation.

Figure 1 shows the discounted realised cashflow at t_{n-1} against the underlying $S_{t_{n-1}}$, with the fitted Laguerre series. We can note that when $S_{t_{n-1}} \lesssim \mathcal{P}$, the discounted cashflow is approximately flat. Economically, the redemption value F dominates because converting yields $NS \leq F$. When $S_{t_{n-1}} \gtrsim \mathcal{P}$, the cashflows increase with S and become increasingly convex. This convexity is the equity-option component embedded in the convertible payoff.

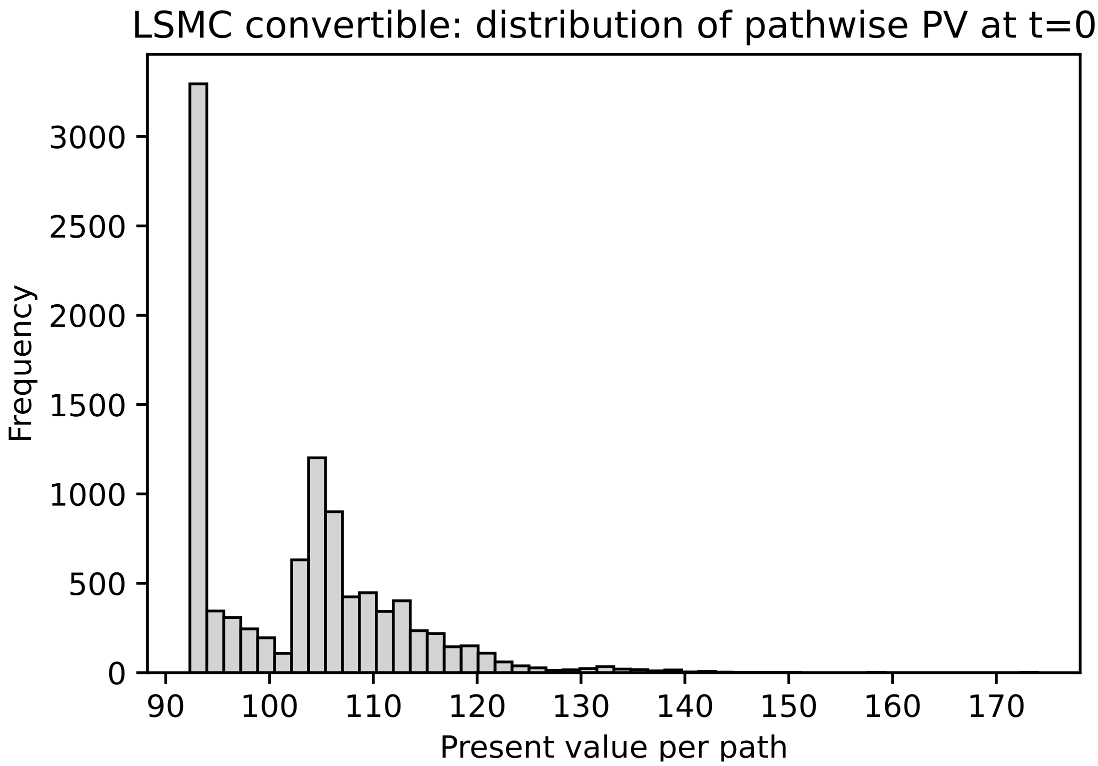

Figure 2: Distribution of pathwise present values at interception. Histogram of pathwise time-0 present values produced by the LSM algorithm. The concentration near the discounted bond floor corresponds to redemption-dominated outcomes, whereas the heavier upper tail corresponds to equity-driven outcomes where conversion becomes optimal and the payoff behaves more like NS_T

The histogram in figure 2 shows the distribution of Monte Carlo pathwise present values $V_0^{(k)}$ returned by the backward induction. A large proportion of paths cluster near a "low" present value level. These scenarios in which the stock does not rise sufficiently above $\mathcal{P}$ and the optimal outcome is essentially redemption at F. The time-0 value in these cases is close to the discounted bond floor Fe^{-rT}. Paths where the stock appreciates substantially lead to conversion and therefore yield larger discounted payoffs that behave increasingly like NS_T. This produced a heavier right tail whereas S_T grew, the payoff became linear in S_T, and the distribution inherited the stock's dispersion.

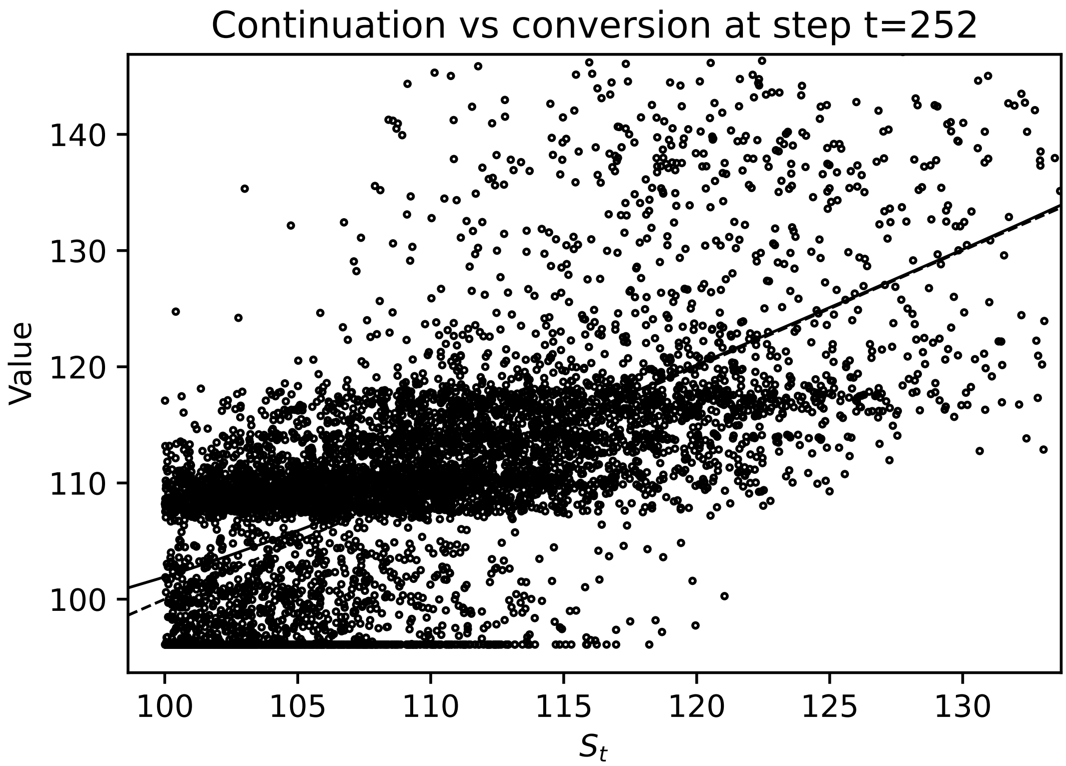

Figure 3: Continuation vs conversion at $t = 252$. Realised discounted hold values plotted against S_t, with the fitted continuation value $\hat{C}(t, S)$ and the immediate conversion value NS. Their intersection defines the estimated conversion threshold S_t^* used by the stopping rule.

We then looked at the realised discounted values against S_t against the continuation curve $\hat{C}(t, S)$ and the immediate conversion value NS, seen in figure 3. The large vertical spread at a given S_t represents that future stock path can diverge substantially, and the terminal outcome depends on that future path and the subsequent optimal stopping decisions. Hence, produces *substantial conditional variance* (heteroskedasticity), typically increasing with S, since higher equity levels amplify the scale of payoff dispersion.

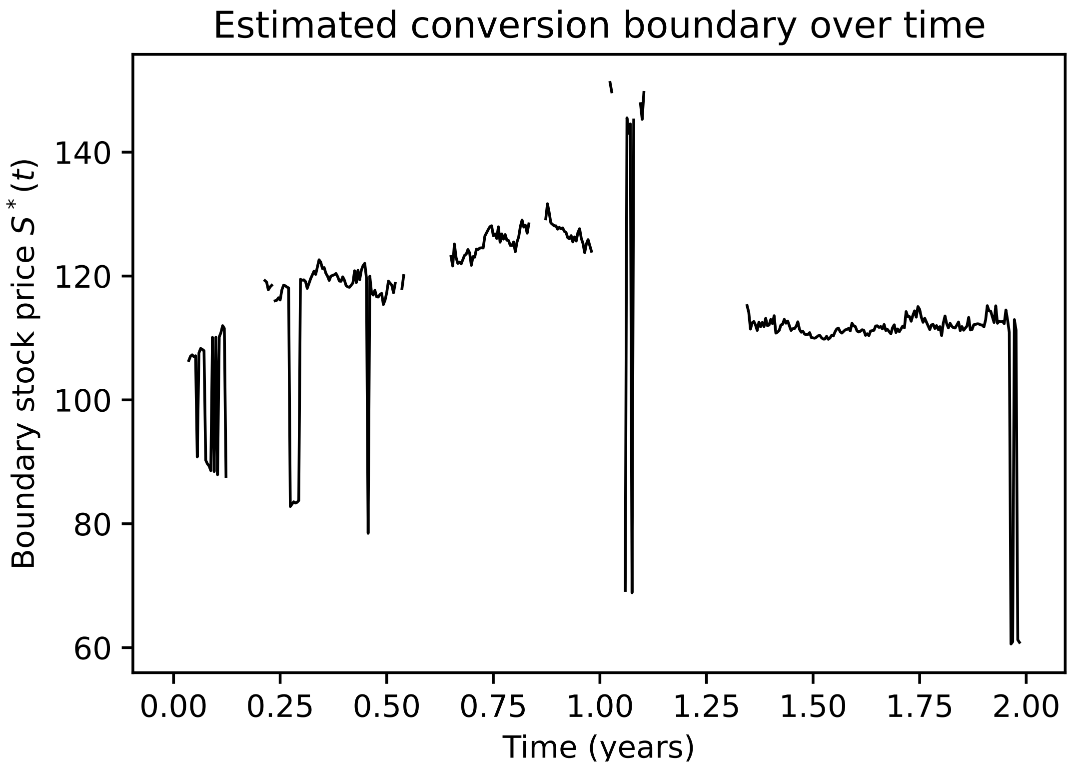

Figure 4: Estimated conversion boundary. Time series of the estimated conversion boundary S_t^* obtained by locating the stock price at which NS first exceeds the fitted continuation value $\hat{C}(t, S)$.

We then estimated the conversion boundary S_t^* extracted across time. Figure 4 shows the conversion boundary would be extremely high, with conversion effectively deferred until maturity for most scenarios, in this frictionless, noise-free idealisation for this specific zero-coupon contract. The estimated boundary is finite and shows noticeable irregularity.

The histogram shown in figure 5 shows the empirical distribution of estimated conversion times $\hat{\tau}$ across simulation paths. The mass at T indicates that the algorithm converts at maturity for most paths, following the no early exercise idea. The smaller amount of prematurity conversions correspond to paths where the regression estimated continuation value falls below NS_T, and hence, triggering early exercise.

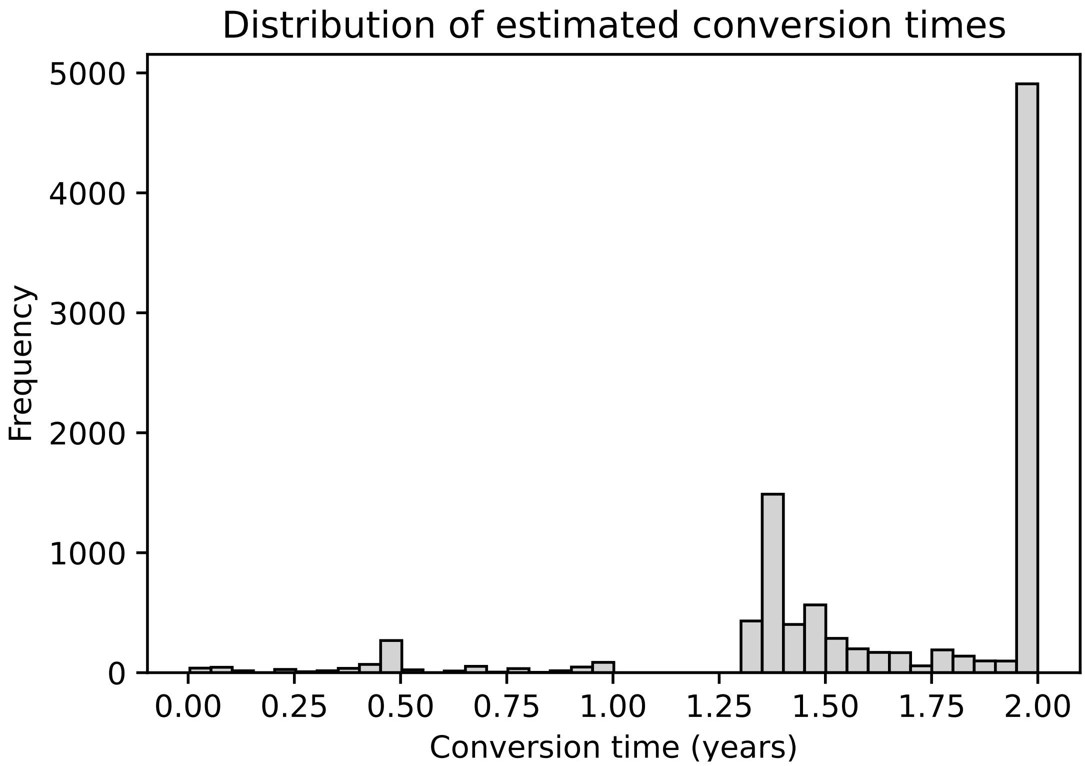

Figure 5: Distribution of estimated conversion times. Histogram of estimated conversion times $\hat{\tau}$ across Monte Carlo paths. The spike at maturity reflects that, for a zero-coupon, zero-dividend contract, optimal conversion is typically deferred until T.

5 Conclusion

In conclusion, we have demonstrated the application of the Longstaff-Schwartz LSMC algorithm to the pricing of convertible bonds. Monte-Carlo approach showed to be a flexible tool for handling the convertible's American-style option whilst capturing the debt-equity like characteristics. We saw that with 1 that the LSM can capture the nonlinearity of the bond reflecting the bond floor when equity is weak, and the convexity when the equity is strong. We then saw that the histogram in 2 showing the convertible bond's payoff decomposition with one regime that represented a debt-like floor, and the other equity-like.Figure 4 tells us where the model is inferring potential early conversion and how stable that inference is over time.

References

Longstaff, Francis A and Eduardo S Schwartz (2001). "Valuing American options by simulation: A simple least-squares approach". In: *The review of financial studies* 14.1, pp. 113–147.